Real-Time Beehive Condition Improving Frequency Selection and Bee Welfare Monitoring and Analysis

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***Abstract*—Beekeeping is a crucial agricultural activity that is needed for pollination and the production of honey. For bee colonies to remain healthy and productive, ideal hive conditions must be maintained. In this study, a real-time hive condition management system is described for beekeepers based on IoT. An ESP8266 microcontroller, a DHT11 sensor for measuring temperature and humidity, and a buzzer are all components of our system. We have collected and evaluated temperature and humidity data from beehives over a long period of time by deploying our monitoring system. We go over our system's practical applications for beekeeping, such as the early identification of unfavorable environmental circumstances and the avoidance of hive stress. We recognize some issues that need additional adjustment, such as sensor accuracy and power supply. This research presents an approach for beekeepers looking to enhance hive management techniques and contributes to the expanding field of IoT in agriculture.**

***Keywords*—Beehive monitoring, Frequency Selection, DHT11 sensor, Blynk cloud, beekeeping, temperature, humidity, hive condition management, pollination services.**

I. INTRODUCTION

Small-scale farmers' daily revenue has been significantly impacted by the meager honey yields from stingless beekeeping. An Internet of Things (IoT)-based monitoring system has been devised to address this problem and increase these beekeepers' revenues. With the help of this method, farmers may learn more about their bee farms and eventually produce more honey. The system entails using a DHT22 sensor and an Arduino Uno ATmega328P to continuously monitor the humidity and temperature within the bee colonies. After gathering, the data is wirelessly sent to a server for tracking and evaluation [1]. In order to provide beekeepers with a comprehensive solution, this work offers a prototype of a real-time monitoring and control system for apiaries [2].

The research presents a multisensory platform that is intended to measure different metrics from field-deployed beehives. It shows how combining information from several sensors can provide understanding of the colony's condition, how it interacts with its surroundings, and how the weather affects it [3]. The authors describe in another paper how they developed a system for remotely monitoring the state of bee colonies. This system was created as a component of the SAMS project for Horizon 2020 and is based on the reasonably priced Wi-Fi microprocessor ESP8266 [4].

Additionally, the authors suggest a novel, reasonably priced device called the Bee Sound Detector (BeeSD), which combines the previously stated sensors with the added capability of real-time sound monitoring for beehive quality management [5]. They intend to give beekeepers an easy-to-use solution by presenting a prototype of an embedded multisensory system for monitoring hives. The efficacy of this small, non-intrusive embedded device in an apiary is demonstrated in the study [6].

Furthermore, a method for widespread, long-term IoT-based bee colony monitoring is proposed. This device is very helpful for field deployment since it can wirelessly send data, identify several features, and is self-sustaining [7]. The paper concludes with a demonstration of an integrated system that combines a multisensory platform for environmental parameter monitoring with video-based honey bee counting. It demonstrates the relationship between honey bee activity and the surrounding environmental factors [8].

II. REVIEW OF LITERATURE

Experts worldwide are becoming increasingly concerned about the depopulation of pollinators. The honeybee (Apis mellifera), one of the most efficient pollinators, stands out among the species in risk of extinction. Numerous reasons, including as varroa mite infestations, heavy chemical usage in agriculture, erratic weather patterns, and diseases particular to bees, are blamed for the rising number of colony collapses [9]. An affordable wireless security system that uses GSM and ZigBee modems to monitor and control unwanted events, such break-ins by robbers or dangerous situations, is one suggested option. To increase the coverage area, cluster heads in various areas connect through GSM with the city's main control center. The PSO algorithm is used to optimize the network for effective routing, and cluster nodes—embedded devices connected to switches and ZigBee modems based on the user's preferences—are installed in each room [10].

Major disasters can result from significant changes in environmental and geographical conditions, needing quick monitoring and responses to preserve lives. The system includes local control rooms, a central control room, and sensor nodes. In order to monitor conditions, sensor data is gathered, and discharge sensors regulate gate openings within predetermined ranges. The central control room, which issues orders to local control rooms depending on the observed conditions, makes all decisions. In the event of severe or abnormal conditions, local control rooms also feature a hooter to signal danger alerts [11-13]. This ground-breaking IoT use case presents real-time monitoring tools for keeping tabs on the number of individuals and alerting users through text message when that number rises above a set threshold. People in rural regions now have a handy way to monitor and react when the number exceeds critical levels thanks to the use of smart IoT-based monitoring devices [14-15].

The development of a bee hive health monitoring system employing image processing methods is described. Two models make up this system: one for detecting bees in photos and another for categorizing the health of the discovered bees [16-18]. In this article, the creation of a system specifically made to photograph bumblebees as they enter and exit their hives from the top and side is described. It demonstrates how these images may be automatically used to follow foraging activities, recognize specific bees, and gauge bee size and pollen presence [19].

For the purpose of gathering environmental and visual data from beehives, an entirely autonomous IoT framework has been created. To identify regions that are under pressure, the framework's services are examined for energy efficiency [20]. Data from numerous sensors that simultaneously measure various beehive parameters are combined on a proprietary platform. The advantages of utilizing more regressors to forecast these variables are shown, and the data is wirelessly transferred to a cloud server [21-26].

III. Block Diagram

The block diagram of Monitoring of Beehive conditions is described in Fig.1. The full beehive monitoring system is represented by the block diagram's core component. This system is made to keep an eye on and control the circumstances inside a beehive, which are essential for the health and productivity of bee colonies. The ESP8266 microcontroller board, also known as NodeMCU, is the central component of this system and acts as its brain. Data collecting, data transmission, and alarm management are just a few of the crucial tasks that the ESP8266 is in charge of.

A DHT11 sensor, which is connected to the ESP8266, is in charge of monitoring the beehive's temperature and humidity levels. Understanding the environmental factors that influence the bees' health and behavior depends on these measures. The ESP8266 receives the data collected by the DHT11 sensor for processing and transmission. Additionally attached to the ESP8266 is a buzzer that acts as an alert system. It can be set to beekeepers' prompt notification of any significant alterations in the beehive's surroundings that may call for attention depending on predetermined conditions or thresholds.

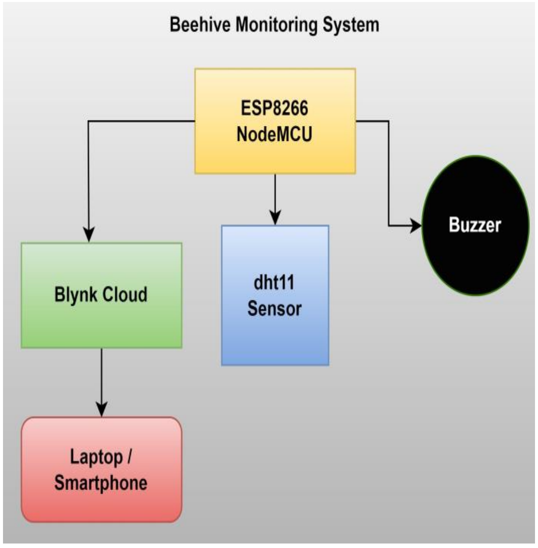

Fig 1. Block diagram of Monitoring of Beehive Conditions

This configuration relies heavily on the Blynk cloud platform, which makes remote monitoring and control possible. The ESP8266 sends the gathered data to the Blynk cloud so that it may be seen and examined there. Through the Blynk app, beekeepers can access this information from their laptops or cellphones, giving them the ability to monitor the health of the beehive even when they are not nearby. Last but not least, the beekeeper's user interface is a computer or smartphone. The Blynk app, which offers real-time data, alarms, and insights regarding the beehive's status, can be accessed and used on these devices. These elements work together to produce a thorough beehive monitoring system management and bee care.

IV. Hardware Development

The hardware interfacing for the Intruder Detection Node is shown in Fig.2. The details as follows:

1) The core of your beehive monitoring system is the ESP8266 microcontroller board. This board serves as the central processing unit and controls all other hardware components in the setup. To connect the ESP8266 to the DHT11 sensor and the buzzer with use jumper wires.
2) The Ground pin of the dht11 sensor has been connected to the common ground and voltage pin

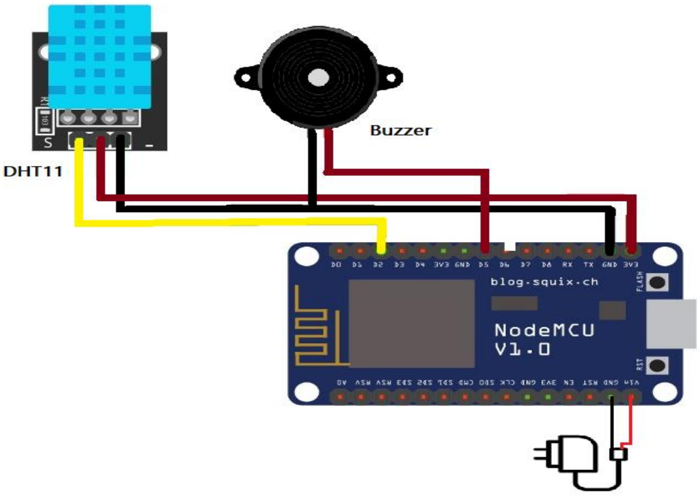

Fig 2. Connection diagram of Monitoring of Beehive Conditions

to the external power. The data pin of the dht11 sensor has been attached with the D5 no pin of the ESP8266 board.

1) The ESP8266 board is linked with a buzzer for alerting purposes. The power pin is linked to the ESP32 board's pin D5, and the buzzer's ground pin is linked to the ground.
2) The ESP8266 board will send all information and alerts to the cloud by help of Wi-Fi.

V. Implementation of the System

The ESP8266 microcontroller board, DHT11 sensor, and Buzzer are initialized to start the system as shown in Fig.3. Make sure the ESP8266 has access to the internet by connecting it to the Wi-Fi network. Continually gather data on temperature and humidity using the DHT11 sensor that was installed inside the hive. The beehive's internal circumstances are represented by this data. Put the gathered information in variables so that it may be processed later. Examine the collected information to find any significant alterations or patterns in the levels of humidity and temperature.

To determine whether the hive's conditions fall within an acceptable range, use predefined thresholds or conditions [27-29]. For instance, confirm that the temperature is within the range that is ideal for bee activity. Set off the buzzer to create an alert if the analysis shows that the conditions have departed from the permitted range. To warn beekeepers of potential problems in the hive, the alert may take the shape of audible signals or patterns. Beekeepers should record the occurrence of each alert along with a timestamp for future use.

Utilize the ESP8266's internet connections to establish a connection with the Blynk cloud computing platform. Send the collected temperature, humidity, and alert status information on a regular basis to the Blynk cloud. This enables remote, real-time hive monitoring by beekeepers. Beekeepers can examine the temperature and humidity data for the hive by logging into the Blynk app on their desktops or cellphones. The program offers visuals to display the data simply and aid beekeepers in seeing trends and patterns, like as graphs or gauges. Set up the Blynk app to notify the beekeeper's device when an alert is generated so that urgent hive conditions can be handled right away. On the ESP8266, keep a local log of the temperature, humidity, and timestamps as shown in Fig.4. This information acts as a backup and is important for historical research.

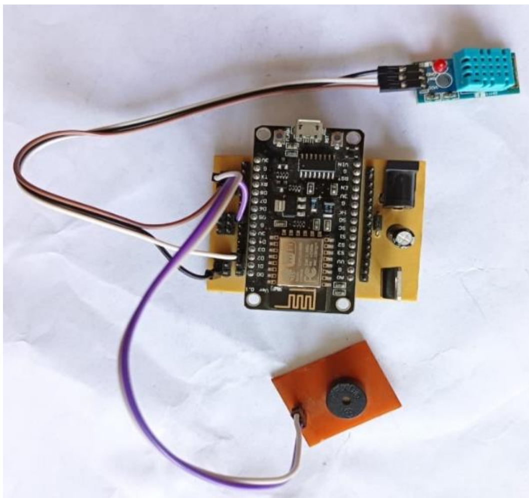

Fig 3. Hardware of Monitoring of Beehive Conditions

The system runs continually, gathering, processing, and transferring data while keeping an eye out for any potential alarms. To guarantee the dependability of data collection and transfer, periodically check for any system defects or connectivity problems. Review the system's efficiency, sensor precision, and power supply on a regular basis to find opportunities for improvement.

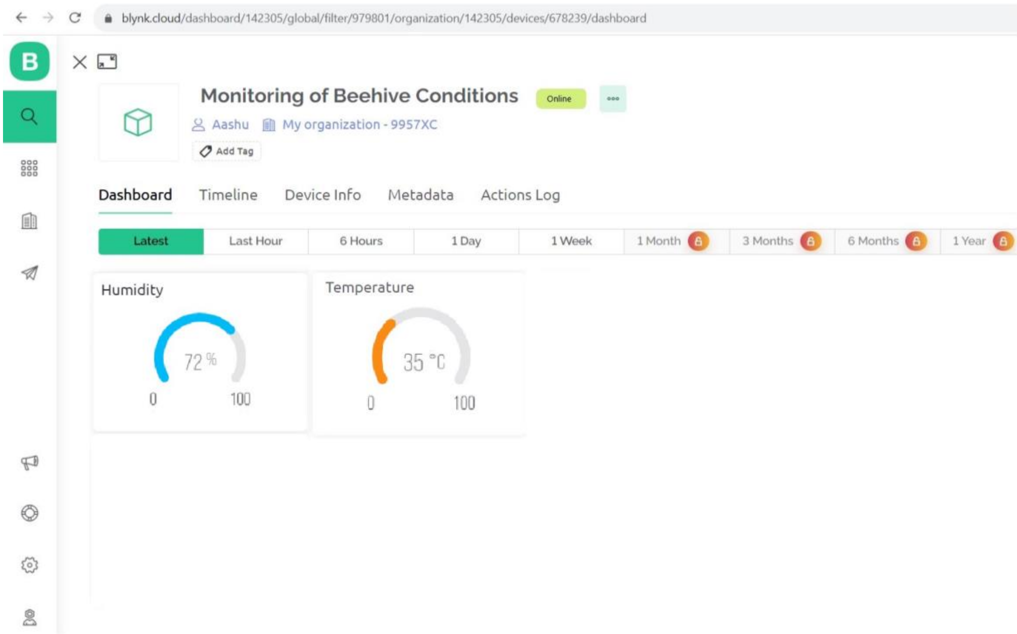

Fig 4. Monitoring of Beehive Conditions on Blynk Cloud Dashboard

Implement improvements based on advice from beekeeping observations and lessons learned. This algorithm describes how beehive monitoring system will work step-by-step, from data collecting inside the hive through remote monitoring and beekeeper alerts. It makes it possible to analyze hive conditions in real-time and makes it easier to act quickly when necessary to protect bee colonies' health and wellbeing [30-32].

VI. CONCLUSION

Therefore, the created home security system in this study, we offer an Internet of Things (IoT)-based Beehive Monitoring System that uses cutting-edge technology to help beekeepers control beehive conditions. The health and productivity of bee colonies depend on maintaining ideal hive conditions, which is a fundamental aspect of beekeeping as an agricultural enterprise. Our system, which is based on the ESP8266 microcontroller, DHT11 sensor, and Blynk cloud platform, provides an efficient method for monitoring beehive data in real-time and granting remote access to it. Our device has proven to have the capacity to offer insightful information on the beehive's environmental parameters through deployment and testing. Beekeepers may remotely check on hive conditions and spot deviations from the ideal range thanks to data on temperature and humidity provided by the DHT11 sensor. The addition of a buzzer for alert generation increases the system's usefulness by informing beekeepers of important changes occurring inside the hive.

The Blynk cloud platform, which offers a user-friendly interface for remote monitoring and control, is essential to our system. Beekeepers have access to the data from their computers or smartphones, giving them the knowledge, they need to manage hives intelligently. Beekeepers can respond quickly to any unfavorable conditions that can endanger the health of their bee colonies because the technology can provide alarms in real-time.

Although we believe that our Beehive Monitoring System has the potential to enhance beekeeping procedures, there are still some obstacles to overcome. The necessity for sensor calibration to increase accuracy and potential power supply optimizations for prolonged deployment periods are two examples of these [33-35]. The system also opens the door for future innovations and data analytics that can aid beekeepers in managing hives even better [36-39]. Our IoT-based Beehive Monitoring System, in conclusion, marks a substantial advancement in beekeeping technology. It gives beekeepers the tools necessary to protect bee health and increase output by giving them immediate access to crucial hive condition data and timely alarms. This technology will benefit beekeepers as well as the crucial pollination services offered by bees in agriculture and ecosystems around the world as we continue to develop and expand it. Beekeeping is positioned for a more sustainable and data-driven future thanks to IoT.

In future, the problem of available bee datasets, which are crucial for state-of-the-art systems and the need for emphasizing on more complex high-quality datasets for developing other applications in this field. The proposed method could be implemented in an embedded device, which could be used for in-field beehive monitoring, or research purposes.

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